

HONORS PHYSICS: ALPINE RESCUE RELAY

HS-PS2-1 • DISTANCE VS TIME

UNIT 2: SCALAR PATH DISTANCE VS. VECTOR DISPLACEMENT • RACE DYNAMICS

Names: _____

Period: 0

Date: _____

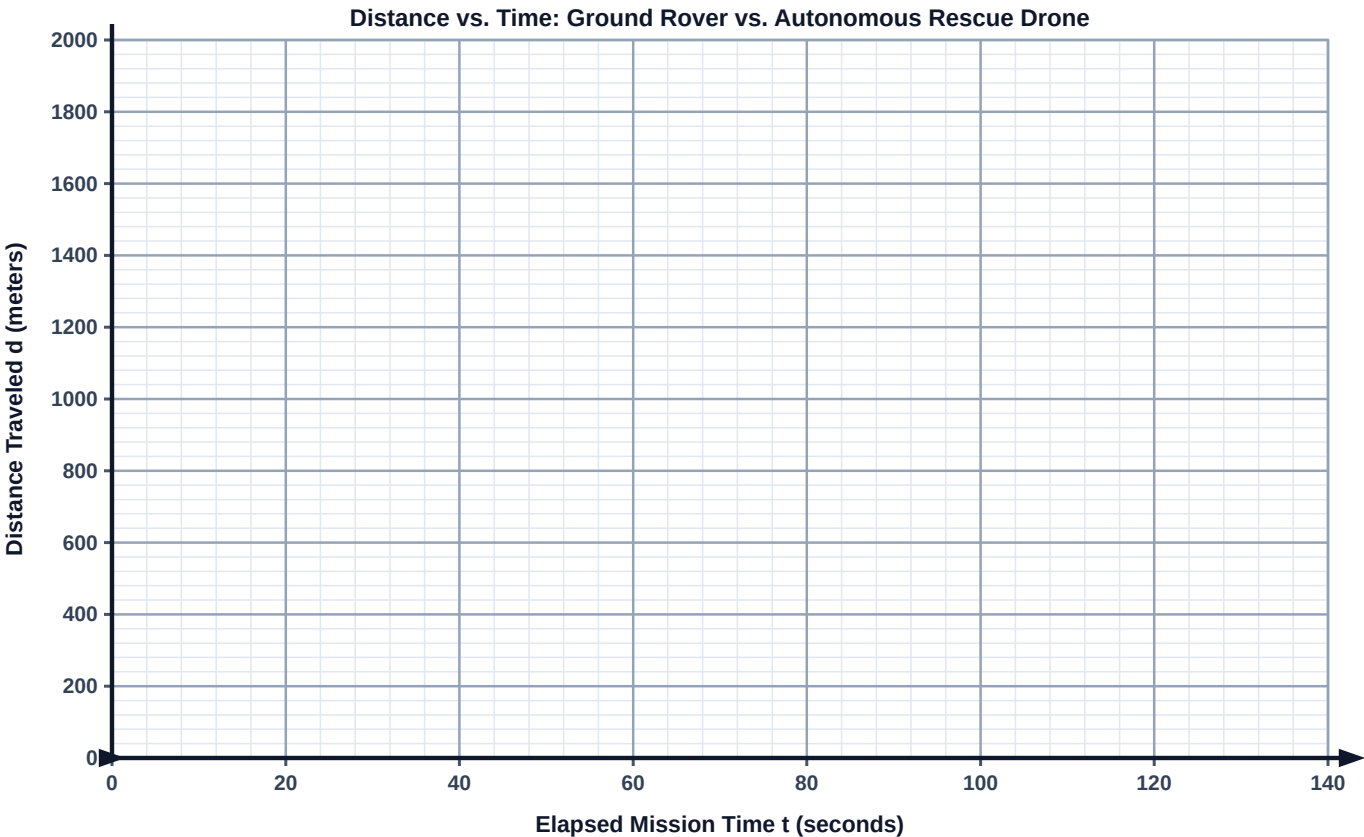
EMERGENCY DISPATCH: ALPINE RELIEF OPERATION

An injured climber activates a beacon at **(800 m East, 600 m North)** from Base Camp (origin = 0 m). Two rescue units depart at **t = 0 s**:

- **Ground Rover Alpha:** Must follow a curved switchback road with a total distance of **1,800 meters** at a steady speed of **15.0 m/s**.
- **Rescue Drone Beta:** Flies straight over the trees. Its straight-line displacement is **1,000 meters** ($\sqrt{800^2 + 600^2} = 1,000$ m). It cruises at **20.0 m/s**, but must land on a ridge for a **35-second thermal cooldown pause** at the 500 m mark (from t = 25 s to 60 s).

 **Graphing Task:** 1) Plot Rover line from (0s, 0m) to (120s, 1800m). 2) Plot Drone: 0-25s (up to 500m), flat line 25-60s (pause), 60-85s (up to 1000m).

RACE
TELEMETRY



GRAPHICAL ANALYSIS & MARGIN OF VICTORY

ANSWER USING YOUR GRAPH ABOVE

1. Destination Arrival Times (From Graph):

At what time does Drone Beta reach its 1,000 m destination? _____ s | At what time does Rover Alpha reach its 1,800 m destination? _____ s

2. Margin of Victory:

Which vehicle reaches the climber first, and by how many seconds? _____ by _____ seconds

3. Path Distance vs. Net Displacement:

Why does the Rover travel 800 meters more distance than the Drone to reach the exact same destination?

4. Maximum Allowable Cooldown (Honors Critical Thinking):

What is the maximum duration the drone could spend cooling down on the ridge and still arrive at the climber before the rover? (Show math):